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Mechanochemical methanolysis of polyethylene terephthalate using calcium oxide as solid base catalyst: a case study†

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Dimethyl terephthalate was obtained in a high yield of 83% from polyethylene terephthalate by planetary ball milling with CaO at ambient temperature as a case study. Although the findings cannot be generalized for a broader set of conditions, the optimal reaction parameters, including the catalyst amount, were found to differ significantly from those for a typical heat-driven reaction.

Plastics have an exceptional range of applications, although it is also vital to reduce the impact of these materials on the environment.¹ This is particularly important as part of a movement to a sustainable global society.² Consequently, there is a requirement to use biomass as a renewable resource and also to reuse waste plastics as an alternative carbon resource.³ The recycling of plastics is essential and chemical recycling is becoming increasingly common in this regard.⁴ In particular, polymers produced by condensation reactions, such as polyesters, can be re-synthesized into high-purity plastics equivalent to those derived from virgin materials by depolymerization followed by recondensation, and can additionally be further upcycled into other value added products.⁵

Polyethylene terephthalate (PET) is a representative polyester and has by far the highest recovery rate among waste plastics. Although this material is already recycled through mechanical processes, these treatments reduce the quality of the product due to degradation reactions associated with repeated use.⁶ Thus, the chemical recycling of PET is of interest and the methanolysis reaction, which is a kind of transesterification using methanol that produces dimethyl terephthalate (DMT) as a monomer, has attracted attention.⁷ Recently, a number of studies have reported the methanolysis of PET using

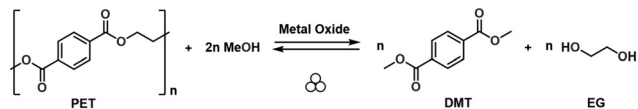
various catalysts.^{8–10} Pham *et al.* obtained a DMT yield of 93% after 24 h of transesterification at room temperature using K₂CO₃ as the catalyst.⁸ The addition of dichloromethane to this process promotes the dissolution and swelling properties of the PET but can result in residual chlorine in the product, which limits practical uses for the material. The use of dichloromethane is being discouraged in industry.¹¹ Tanaka *et al.* examined the reaction of dimethyl carbonate with ethylene glycol, which is generated during transesterification along with DMT, to produce ethylene carbonate and methanol.⁹ This process removed ethylene glycol from the reaction system to shift the equilibrium and increase the DMT yield. This reaction proceeded rapidly at room temperature when using LiOMe as a homogeneous catalyst. This is an excellent process for obtaining high yields of DMT by filtration, evaporation, and water washing after the reaction. A similar system involving the reaction of ethylene glycol with allylboronic acid to produce useful compounds was recently reported by Zhang *et al.*,¹⁰ who employed calcined Mg–Al hydrotalcite, a layered double hydroxide, as a reusable heterogeneous base catalyst, although a high reaction temperature of 180 °C was required.

Mechanochemical reactions have attracted attention in recent years as environmentally friendly processes.¹² These reactions proceed on the basis of mechanical energy generated by contact with solid spheres, unlike typical processes that require heating. This approach is especially useful in the case that solid reactants and solid catalysts are used, because the good solid-to-solid contact in such systems significantly promotes reactivity.¹³ Mechanochemical processing has been applied to biomass conversion using solid catalysts, including the hydrolysis of cellulose, which is a key reaction associated with the utilization of lignocellulosic biomass.^{14–16} In addition, the depolymerization of plastics (including PET) via mechanochemical reactions has been widely investigated.^{17–19} Štrukil demonstrated near-qualitative PET conversion and high disodium terephthalate yields based on mechanochemical milling with NaOH in a mill.¹⁷ In this prior work, efficient solid-state alkaline hydrolysis of PET was achieved at ambient temperature and pressure

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Scheme 1 Mechanochemical methanolysis of PET using metal oxide catalyst.

without the use of an organic solvent. Tricker *et al.* examined the mechanism associated with the mechanochemical depolymerization of PET with NaOH.¹⁸ Tricker's work indicated that the monomer yield was proportional to the ball-to-reactant mass ratio, indicating that collision energy promoted the reaction. While the hydrolysis of PET to terephthalic acid using mechanochemical methods has been reported, methanolysis to DMT has not yet been demonstrated. On this basis, the present work examined the methanolysis of PET *via* a mechanochemical process employing solid catalysts at ambient temperature and pressure (Scheme 1). This process was found to generate a high yield of DMT (83%) after 4 h with the addition of a small amount of CaO (6 wt%). The optimal reaction conditions were investigated and the relationship between reactivity and the applied impact energy was assessed.

Fig. 1(a) summarizes the DMT yields obtained by ball milling with the addition of various metal oxides. Note that no DMT was produced without the addition of an oxide. Similarly, no or only minimal DMT was obtained with the acidic oxides MoO₃, WO₃, Ta₂O₅, Nb₂O₅ and TiO₂. A small amount of DMT was formed in the case of the amphoteric oxide ZnO, while the formation of DMT was significant over the basic oxides MgO and CaO. A particularly high yield of DMT (75%) was obtained with CaO, without the formation of other products such as 2-hydroxyethyl methyl terephthalate. No DMT was produced over SrO which was dissolved after the reaction and instead only a small amount (*ca.* 12%) of monomethyl terephthalate was obtained when using strongly basic oxide BaO. Other calcium compounds, including Ca(OH)₂, CaCO₃ and Ca(OMe)₂, were also tested. The hydroxide Ca(OH)₂ showed the second highest activity after CaO, whereas CaCO₃ and Ca(OMe)₂ also produced DMT but in lower yields.

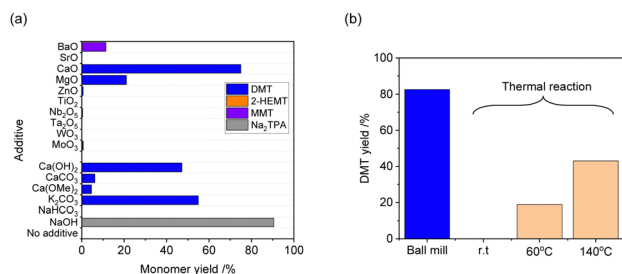


Fig. 1 (a) Catalyst screening for mechanochemical methanolysis of PET using planetary ball mill. Reaction conditions: PET (0.15 g), additive (0.15 g), methanol (1.5 mL), 600 rpm, 6 h. 2-HEMT: 2-hydroxyethyl methyl terephthalate. MMT: monomethyl terephthalate. (b) Comparison of activity during mechanochemical methanolysis using a planetary ball mill and conventional thermal methanolysis in a pressure-resistant stainless steel tube reactor. Reaction conditions: PET (0.15 g), CaO (10 mg, 6 wt%), methanol (1.5 mL), 4 h.

K₂CO₃, which was used in a previous study⁸ and found to function as a highly active solid base catalyst for the methanolysis of polycarbonates,²⁰ also produced DMT. However, the resulting yield was lower than that provided by CaO under the present reaction conditions. In addition, although NaOH is known to be highly active with regard to the conversion of PET,^{17,18} the sodium salt of terephthalic acid was produced rather than DMT when using this base. The results showed that the reaction did not proceed solely on the basis of the impact energy generated by the contact between the balls or between the balls and the vessel walls (that is, in the absence of a catalyst). Hence, the methanolysis reaction only proceeded efficiently in the case that a basic solid catalyst such as CaO was included.

Fig. 1(b) compares the results for the mechanochemical reaction with those for conventional heat-driven reactions. In these trials, thermal methanolysis was performed using a pressure-resistant stainless steel tube reactor with a polytetrafluoroethylene inner tube, employing different temperatures but otherwise using the same conditions as in the mechanochemical process. Interestingly, the reaction did not proceed at all at room temperature and the DMT yield was only 43% even at 140 °C, meaning that the efficiency was much lower than that obtained by ball milling. The container may have been hot during the ball milling process and so temperature labels, which irreversibly changed colour with variations in temperature, were affixed to the lid and base of the zirconia vessel so as to ascertain the temperature during the mechanochemical reaction. This test indicated that the temperature of the vessel remained below 40 °C throughout the reaction (Fig. S1, ESI†). Therefore, at least from a macroscopic viewpoint, the reaction temperature during the ball milling process was low and so the reaction was evidently driven by mechanical forces. For organic mechanochemical reactions, *in situ* measurements using a combination of Raman spectroscopy and thermography indicated that the temperature increase was close to 10 K.²¹ For inorganic reactions, a huge rise to 2000 K that triggers ignition was directly visualized.²² It should be noted that this measurement does not deny the hot spot theory that the local temperature increase is one of the key features of mechanochemistry.

The mechanochemical reaction with CaO was subsequently optimized by investigating the effects of varying the amounts of methanol, CaO and PET on the DMT yield. The relationships between these variables and the DMT yield are shown in Fig. 2(a), 2(b) and 2(c), respectively. Methanol was the reaction reagent and so naturally no DMT was obtained in the absence of methanol (Fig. 2(a)). The DMT yield initially increased with increasing amount of methanol up to 1.5 mL, above which the DMT yield decreased. In a typical thermal reaction, the DMT yield would be expected to increase along with the amount of methanol, but this was not the case in the present reaction system. This difference is attributed to the fact that the reaction was driven by collision energy and mechanical energy was not transferred to the PET and catalyst powder as effectively at higher methanol loadings. Unexpectedly, the highest DMT yield (83%) was obtained with a very small amount (10 mg, 6 wt%) of

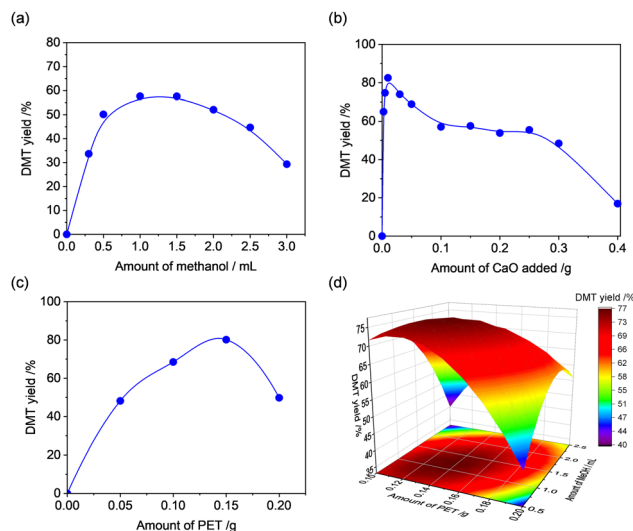


Fig. 2 Results for optimization of amounts of methanol, CaO and PET using rotational rate of 600 rpm for 4 h. (a) The effect of the amount of methanol with PET (0.15 g), CaO (0.15 g) and methanol (0.5–3.0 mL), (b) the effect of the amount of CaO with PET (0.15 g), CaO (2.5 mg–0.40 g) and methanol (1.5 mL) and (c) the effect of the amount of PET with PET (0.05–0.20 g), CaO (10 mg) and methanol (1.5 mL). (d) The response surface for DMT yield using 10 mg CaO.

CaO (Fig. 2(b)). As noted, the reaction did not proceed without the addition of a catalyst and the turnover number was calculated using the amount of base of the CaO (0.15 mmol g^{-1}) determined by titration (Table S1, ESI†). This value was found to be 540, indicating that the reaction proceeded very efficiently. In a standard heat-driven reaction, a greater amount of catalyst will generally increase the activity. However, in the present reaction system, the DMT yield increased when using 2.5 and 5 mg CaO, with the highest yield at 10 mg, followed by a decrease in yield at higher amounts. These results suggest that solid–solid contact was an important aspect of this reaction, as was also found when varying the methanol content. Fig. 2(c) shows the relationship between the amount of PET and the DMT yield. If the reaction rate was proportional to the amount of substrate, meaning if this was a first-order reaction, then increasing the amount of PET would not change the DMT yield. However, in the present experiments, increasing the amount of PET raised the DMT yield. In addition, the DMT yield decreased when the amount of PET exceeded approximately 0.15 g. Note that, while DMT yields are plotted here, the same trend was observed when plotting the absolute amount of DMT (Fig. S2, ESI†). The total volume was calculated using six balls, methanol, CaO, and PET. The free space in the case of 0.15 g and 0.20 g PET was estimated to be 60.2% and 60.0%, respectively. Therefore, this is unlikely to be the cause of the sharp drop in yield, but rather the limitation of impact energy due to the lower ball to PET ratio.

A response surface methodology process based on a central composite design using three factors (the amounts of CaO, PET and methanol) was used to further analyze the interrelationship between these parameters and the DMT yield. The experimental design is summarized in Table S2 (ESI†) while Fig. 2(d)

presents the results. Similar to the results for each factor, the highest DMT yields were obtained with approximately 0.15 g PET and 1.5 mL methanol, with sharp decreases in the yield at higher amounts. The *p* values obtained from ANOVA calculations demonstrated significant interactions between the amounts of CaO and methanol and between the amounts of PET and methanol, and not the independent factor (Table S3, ESI†). This outcome is in good agreement with the results showing that physical contact between the PET and solid catalyst was an important driver for the mechanochemical reaction, and that this contact was greatly affected by the amount of methanol.

Fig. S3 (ESI†) plots the reaction yields over time and also provides scanning electron microscopy (SEM) images of the PET powder before and after the ball milling. As noted above, a higher DMT yield was obtained with 0.010 g CaO compared with 0.150 g. It is apparent that the DMT yield increased over time in an almost linear fashion at the start of the reaction, and that the reaction rate increased substantially at the 2 h mark. The SEM images show that the PET particles had been ground into a fine powder at this point, and so the reaction rate presumably increased because the finer particles made better contact with the CaO and methanol. After a reaction time of 4 h, the DMT yield was either constant or slightly decreased, presumably due to the backward reaction of the DMT with ethylene glycol. An additional experiment confirmed that DMT was consumed during ball milling with CaO in methanol in the presence of ethylene glycol (Fig. S4, ESI†). Mechanochemical processing is known to promote the depolymerization of PET²³ and cellulose¹⁵ but also to drive various recondensation reactions. Another reason for the constant yield of DMT after 4 h is the formation of calcium terephthalate which was confirmed by XRD measurement (Fig. S5, ESI†).²⁴ Recycling of the CaO catalyst was also examined and, although the DMT yield was found to decrease by approximately 30% after three uses of the same material, the yield plateaued after that point (Fig. S6, ESI†).

Fig. 3 shows the effects of various each ball milling conditions on the DMT yield. As the number of balls in the vessel was increased from one to six, the DMT yield increased monotonically (Fig. 3(a)). In other trials, balls having 10, 5 and 3 mm diameters were employed (Fig. 3(b)). In these experiments, 46 balls of 5 mm diameter or 213 balls of 3 mm diameter were used to achieve the same total mass as 6 balls of 10 mm diameter, and the DMT yield was highest with the 10 mm balls. Lastly, the activity was accessed while varying the rotational rate from 200 to 800 rpm (Fig. 3(c)). At 200 rpm, the DMT yield was low (less than 15%) whereas at 300 rpm the yield exceeded 50% and a maximum yield of 83% was observed at 600 rpm. At higher rotational rates (700 and 800 rpm), the DMT yield decreased.

Very recently, Jafer *et al.* calculated the impact energy generated by planetary ball mills and shaker mills based on kinematics and correlated these theoretical results with various reaction activities.²⁵ Using the publicly available energy calculation tool developed by Jafer,²⁶ the impact energy values under the present reaction conditions were ascertained. The total impact

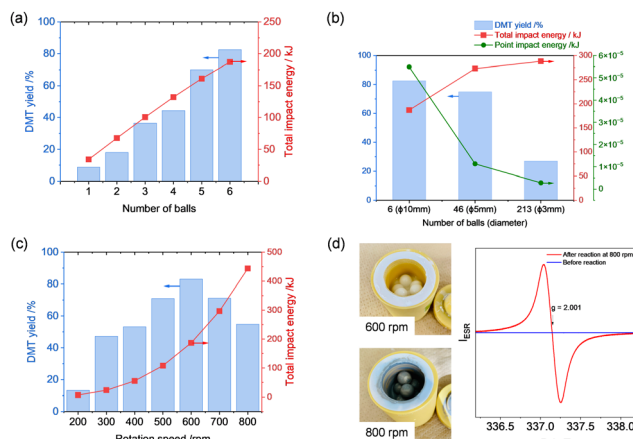


Fig. 3 Effects of (a) number of balls, (b) ball size (keeping total mass constant) and (c) rotational rate on DMT yield. (d) Photographic images of reaction vessel after trials at 600 and 800 rpm, and ESR spectra obtained from PET before reaction and from solid mixture after reaction at 800 rpm.

energy was found to increase according to either a linear or a quadratic relationship with increasing number of balls and rotational rate, respectively. The reactivity was also determined to be highly correlated with the impact energy. At the higher rates of 700 and 800 rpm, the DMT yield decreased while the solid powder produced by the reaction was grey in colour (Fig. 3(d)). It is likely that overly high impact energies promoted reactions other than methanolysis, such as oxidative radical decomposition of the PET.²³ In fact, the formation of organic radicals during ball milling was suggested by electron spin resonance (ESR) analyses (Fig. 3(d)).²⁷ During the trials using different ball sizes with a constant total mass (Fig. 3(b)), the larger balls provided greater activity even though the total impact energy was higher in the case of smaller ball sizes. Despite this, the instantaneous impact energy, meaning the impact energy at any given point in time, was higher for the larger balls, which was consistent with the activity trend.

To summarize, the methanolysis of PET in a planetary ball mill at ambient temperature and pressure with 6 wt% CaO provided a high yield of DMT (83%). This reaction uses only a minimal amount of methanol, does not require any other organic solvents, and the methodology using mechanochemical depolymerization could be adapted to allow the treatment of other polymers.

AT: conceptualization; SC and AT: investigation and methodology; SC and AT: formal analysis; AT: supervision; AT: funding acquisition; SC and AT: writing – original draft; AT: writing – review & editing.

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Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

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